

TUTORATO ANALISI I

06/10/'26

- proprietà delle potenze
dei logaritmi.
- operazione con grafici $f(x) \rightsquigarrow f(1/x)$
 $|f(x)|$ ecc.
- iniettività, suriettività, ...

Proprietà delle potenze

$$a, b \in \mathbb{N}_{>0} \quad a^b = \underbrace{a \cdot \dots \cdot a}_b$$

$$\text{I) } a^b \cdot a^c = a^{b+c}$$

$$\underbrace{(a \cdot \dots \cdot a)}_b \cdot \underbrace{(a \cdot \dots \cdot a)}_c = \underbrace{a^{b+c}}_{b+c} \quad \checkmark$$

$$\text{II) dati: } a, b, c \in \mathbb{N}_{>0} \quad c < b$$

$$a^b / a^c = a^{b-c}$$

$$\frac{\underbrace{a \cdot \dots \cdot a}_{b-c} \cdot \cancel{a \cdot \dots \cdot a}_c}{\cancel{a \cdot \dots \cdot a}_c} = a^{b-c} \quad \checkmark$$

Se voglio estendere a esponenti negativi,
date I) e II), ho un solo modo di farlo.
In particolare, deve valere

$$a^{-c} = \frac{1}{a^c}$$

$$\Leftrightarrow a^b \cdot a^{-c} = a^b / a^c$$

$$\Leftrightarrow a^{b-c} = a^{b-c} \quad \checkmark$$

III) $\forall a, b, c \in \mathbb{N}_{>0} \quad (a^b)^c = a^{b \cdot c}$

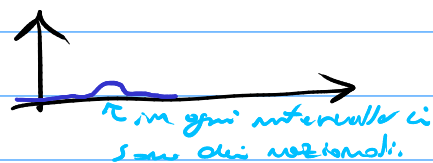
$$c \left\{ \begin{array}{l} \overbrace{(a \cdot a \cdot \dots \cdot a)}^b \cdot \\ (a \cdot a \cdot \dots \cdot a) \cdot \\ \dots \dots \dots \\ (a \cdot a \cdot \dots \cdot a) \end{array} \right. = (a^b)^c$$

Nel rettangolo ci sono $b \cdot c$ "a"

$$\Rightarrow (a^b)^c = a^{b \cdot c} \quad \checkmark$$

FUN FACT:

$f: \mathbb{R} \rightarrow \mathbb{R}$ continua
è univocamente determinata
dalla sua restrizione a $\mathbb{Q}$,
cioè dai valori che assume
sui razionali.



⚠ $(a^b)^c \neq a^{b^c}$

ES: $(3^3)^3 = 3^{3 \cdot 3} = 3^9$
 $3^{3^3} = 3^{27} \neq 3^9$

Se vogliamo estendere a esponenti anche razionali $\frac{p}{q}$, di nuovo, se vogliamo che III) continui a valere, possiamo farlo in un solo modo.

$$(3^{1/2})^2 \stackrel{\text{IV}}{=} 3^{\frac{1}{2} \cdot 2} = 3 \\ = \sqrt[2]{3}.$$

In generale $a^{1/q} = \sqrt[q]{a}$.

$$3^{5/2} = (3^{1/2})^5 \quad a^{p/q} = (a^{1/q})^p = (\sqrt[q]{a})^p.$$

$$\left(\sqrt[17]{2^5} \right)^{11} = \left((2^5)^{1/17} \right)^{11} = \left(2^{5/17} \right)^{11} = 2^{\frac{5}{17} \cdot 11} = 2^{55/17}.$$

$$\left(((a^b)^c)^d \dots \right)^z = a^{b \cdot c \cdot \dots \cdot z}.$$

Proprietà dei Logaritmi

definisce ^{per $x > 0$} $\log_a(x)$ come quel numero reale

tale che $a^{\log_a(x)} = x$.

ESEMPIO: $\log_2(a) = 1$
 $\log_2(a^b) = b$

Valgono "di nuovo" I, II e III viste da una prospettiva diversa.

I) $\log_a(xy) = \log_a(x) + \log_a(y)$

$$\begin{aligned}
 (\Rightarrow) \quad a^{\log_a(xy)} & \stackrel{?}{=} a^{\log_a(x) + \log_a(y)} \\
 & = \quad \quad \quad = \text{I)} \\
 xy & \quad \quad \quad \underbrace{a^{\log_a(x)}}_x \cdot \underbrace{a^{\log_a(y)}}_y
 \end{aligned}$$

$$\text{II)} \quad \log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y \quad \text{es: } \log\left(\frac{1}{x}\right) = -\log x.$$

$$\begin{aligned}
 (\Rightarrow) \quad a^{\log_a\left(\frac{x}{y}\right)} & = a^{\log_a(x) - \log_a(y)} \\
 & = \quad \quad \quad = \text{II)} \\
 x/y & = \frac{a^{\log_a(x)}}{a^{\log_a(y)}}
 \end{aligned}$$

$$\text{III)} \quad \log_a(x) = \frac{\log_a(x)}{\log_a(a)}$$

$$\text{es: } a = a^1$$

$$\log_a x = \frac{1}{2} \log_a x$$

$$(\Rightarrow) \log_a(a) \cdot \log_a(x) = \log_a(x)$$

$$\text{REMEMBER: } (a^b)^c = a^{b \cdot c}$$

$$\begin{aligned}
 (\Rightarrow) \quad a^{\log_a(a) \cdot \log_a(x)} & = a^{\log_a(x)} \\
 & = \text{III)} \\
 & = \underbrace{(a^{\log_a(a)})}_{a}^{\log_a(x)} \\
 & = a^{\log_a(x)} \\
 & = x
 \end{aligned}$$

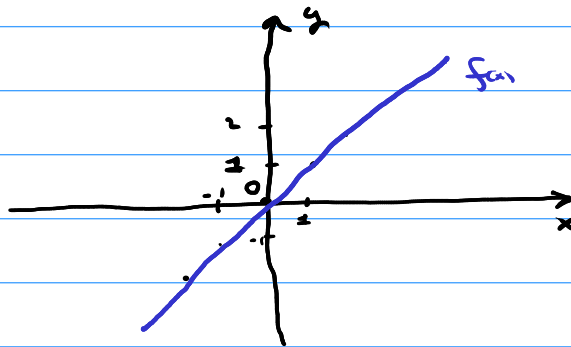
GRAFICI

ESEMPI:

(A)

$$f: \mathbb{R} \rightarrow \mathbb{R} \quad (\dots \text{ossia } f(x) = x)$$

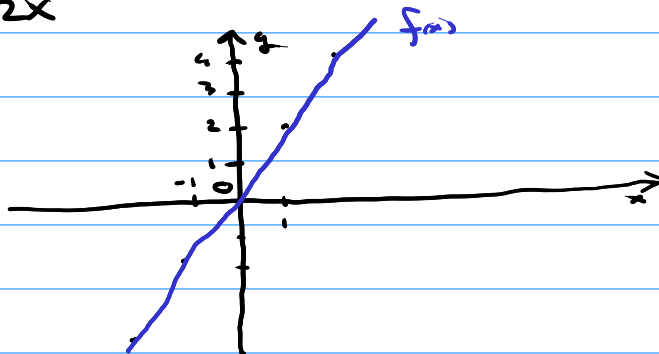
$$\underline{x \mapsto x}$$



(B)

$$f: \mathbb{R} \rightarrow \mathbb{R} \quad (\dots \text{ossia } f(x) = 2x)$$

$$x \mapsto 2x$$

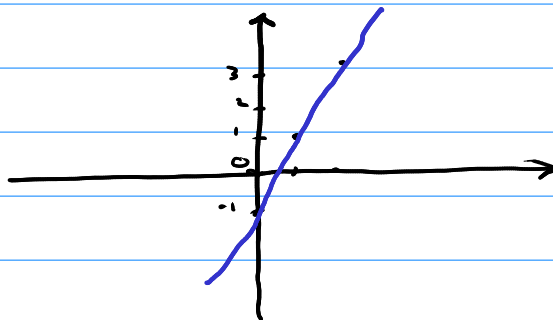


(C)

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto 2x - 1$$

$$(\dots \text{ossia } f(x) = 2x - 1)$$

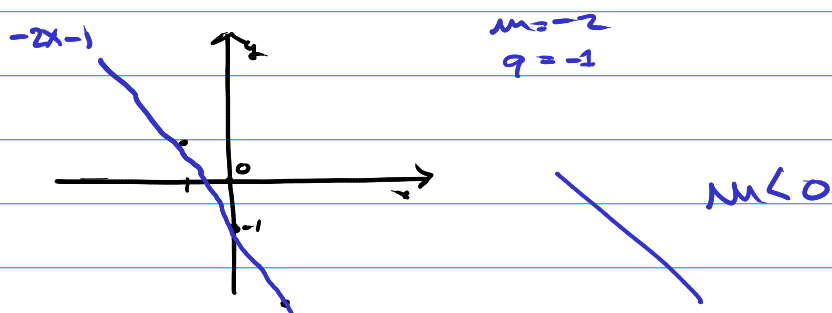
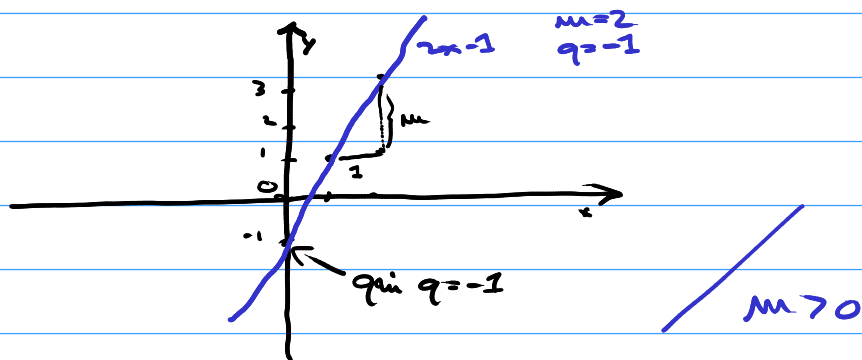


Per le prossime volte:

- funzioni goniometriche
- recap derivate & integrali
- disequazioni & diseg.
irrazionali: (e.g. $\sqrt{f(x)} > g(x)$).
- valore assoluto (grafici, deriv...)

(D) $f: \mathbb{R} \rightarrow \mathbb{R}$
 $x \mapsto mx + q$

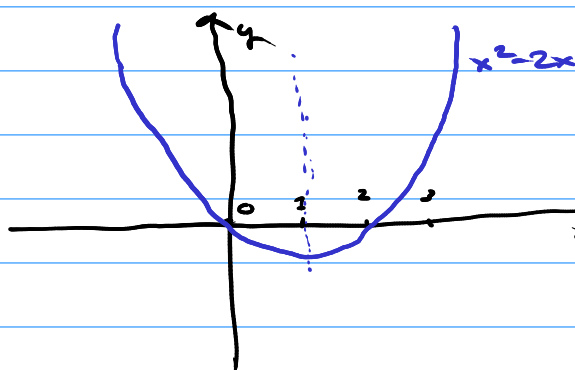
S. $f(0) = m \cdot 0 + q$
 $f(0) = q$



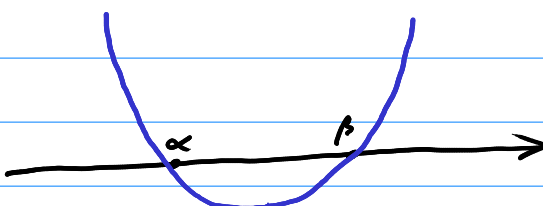
(E) $f: \mathbb{R} \rightarrow \mathbb{R}$
 $x \mapsto x^2 - 2x$
 $x(x-2)$

(... also: $f(x) = x^2 - 2x$)

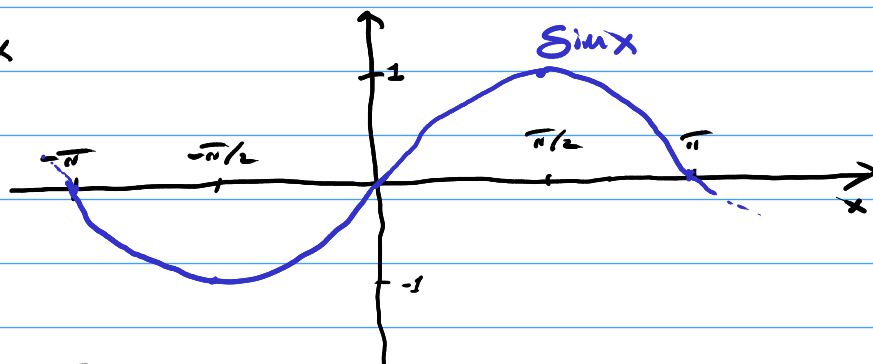
Oss: $x(x-2)$ is
 $= 0$ for $x \in \{0, 2\}$
 < 0 for $0 < x < 2$
 > 0 for $x < 0$ or $x > 2$



(F) $f: \mathbb{R} \rightarrow \mathbb{R}$
 $x \mapsto (x-\alpha)(x-\beta)$ can $\alpha < \beta$

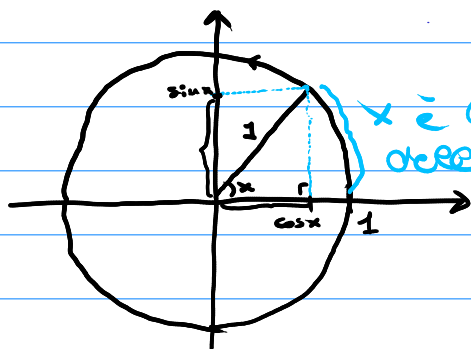
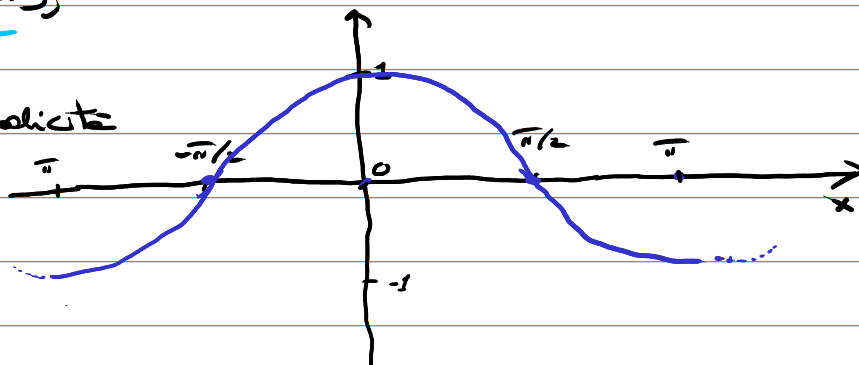


⑨ $f(x) = \sin x$



$\sin x$ e $\cos x$ sono
definiti per $x \in [0, 2\pi]$,
 $0 \leq x \leq 2\pi$

poi si estende per periodicità
con periodo 2π .



x è la lunghezza
dell'arco di circonferenza.

$$\sin^2 x + \cos^2 x = 1^2.$$

(=1)